PHARMAFLOW INTELLIGENCE

REPORT TYPE
Comprehensive Repurposing Scan

MOLECULE
Doxycycline

DATE
2026-02-24

Doxycycline

Strategic Analysis & Market Viability Report

Confidential & Proprietary
Generated by PharmaFlow AI System

Executive Briefing

MARKET MATURITY

Mature

VENDOR AVAILABILITY

5 Public Vendors

PATENT STATUS

Doxycycline is currently generic, as its patent expired many years ago.

Key Findings

Summary: Doxycycline is a mature generic antibiotic with a long history of use. Its patent has expired, and it is widely available in various formulations and strengths from multiple manufacturers. The drug has been extensively studied and utilized for the treatment of various infections. Currently, there are two active clinical trials investigating the use of Doxycycline in different conditions, including COVID-19, infection, chronic endometritis, relapsing fever, and intrauterine adhesion. The presence of these trials suggests that researchers are still exploring the potential of Doxycycline in new therapeutic areas. Additionally, there have been five recent publications related to Doxycycline, indicating ongoing interest in the scientific community. However, as a generic drug, Doxycycline faces significant challenges in terms of differentiation and market competition. Any potential strategy for repurposing or repositioning Doxycycline would need to take into account the existing market landscape and the need for new intellectual property (IP) to create a competitive advantage.

Strategic Drivers

Repurposing potentials

Critical Risks

Regulatory changes

Market Intelligence

Market Structure

> Doxycycline is a mature generic drug, with its patent having expired long ago. It is widely available in various formulations and strengths from multiple manufacturers.

Key Global Manufacturers

Major players identified in this therapeutic area:

- Pfizer
- Mylan
- Teva Pharmaceutical Industries
- Sandoz
- Akorn

Scientific Literature Review

Key Publications

The following peer-reviewed articles were identified as high-relevance for the repurposing hypothesis:

Chelating anticoagulants reduce plasma doxycycline measurements: implications for doxy PEP monitoring.

Expert review of clinical pharmacology • 2026 • PMID: 41701614

"

Background
Doxycycline post-exposure prophylaxis is recommended for men who have sex with men at high risk for sexually transmitted infections. Accurate doxycycline measures ensure..."

Ivermectin and Doxycycline combination as a promising anti-viral drug candidate: an in-silico and DFT study.

In silico pharmacology • 2026 • PMID: 41717425

"In contemporary area of medical research, repurposing of drugs has emerged as a promising strategy in drug discovery against viral infections in individual or in combination modes...."

Trends in Chlamydia Prescribing Practices After Changes to National Treatment Guidelines in the United States, 2016-2023.

Sexually transmitted diseases • 2026 • PMID: 41082782

"

Background
In 2021, the US Centers for Disease Control and Prevention revised Chlamydia trachomatis (CT) treatment guidelines to recommend only doxycycline as first line-therapy, instead..."

Timing and efficacy of doxycycline in macrolide-resistant Mycoplasma pneumoniae pneumonia in children: a single-center retrospective study.

Frontiers in public health • 2026 • PMID: 41717612

"

Background
Outbreaks of macrolide-resistant Mycoplasma pneumoniae (MRMP) in children have posed ongoing treatment challenges. This study aimed to assess the efficacy of doxycycline..."

Post-Exposure Prophylaxis Prescribing Practices in a Lyme Disease-Endemic Area.

Infectious disease reports • 2026 • PMID: 41718073

"

Background/objectives
The 2020 Infectious Diseases Society of America (IDSA) guidelines recommend a single 200 mg dose of doxycycline within 72 h of tick removal after a high-risk bite for..."

Clinical Development

Active & Recruiting Trials

STUDY TITLE	NCT ID	PHASE	STATUS	CONDITIONS
Intralesional Voriconazole, or Intralesional Cryotherapy, or Oral...	NCT05708625	PHASE3	UNKNOWN	Leishmaniasis, Cutaneous
Doxycycline to Upgrade Organ Response in Light Chain (AL) Amyloidosis Trial	NCT02207556	PHASE2	COMPLETED	Primary Systemic Amyloidosis
Comparison Between aPDT and Systemic Doxycycline on Non-surgical...	NCT01595594	PHASE3	COMPLETED	Periodontal Disease, Type 2 Diabetes
Azithromycin Compared With Doxycycline for Treating Anorectal...	NCT03532464	PHASE4	UNKNOWN	Chlamydia Trachomatis Infection,...
Doxycycline for Elbow Tendinopathy	NCT04686799	PHASE1	ACTIVE_NOT_RECRUITING	Lateral Epicondylitis, Lateral...
Omega-3 Pufa and Aspirin as Adjunctive Therapy in Treatment of...	NCT07385300	NA	COMPLETED	Periodontitis, Diabete Mellitus
Doxycycline in Treatment of Bleeding With Depot...	NCT01254799	PHASE3	COMPLETED	Uterine Hemorrhage
The Effect of Doxycycline on Matrix Metalloproteinase Expression...	NCT00538967	PHASE2	COMPLETED	Aortic Aneurysm, Abdominal
A Pilot of Pediatric/Adult Study of Gene Expression Profiling and...	NCT00353158	PHASE1	COMPLETED	Healthy Volunteers, Fungal...
A Prospective Trial of Nasal Mupirocin, Hexachlorophene Body Wash,...	NCT01049438	NA	COMPLETED	Methicillin Resistant Staphylococcus...

RepurpoSing Old Drugs TO SuppRess a Modern Threat: COVID-19 STORM	NCT04433078	PHASE2	WITHDRAWN	Cytokine Storm, SARS-CoV-2
Non-Invasive Treatment of Abdominal Aortic Aneurysm Clinical Trial	NCT01756833	PHASE2	COMPLETED	Aneurysm
Effectiveness of Ivermectin and Doxycycline on COVID-19 Patients	NCT04591600	PHASE1, PHASE2	COMPLETED	Covid19
Single Dose Azithromycin in the Treatment of Adult Cholera	NCT00229944	PHASE3	COMPLETED	Cholera
Development of an Optimal Antibiotic Regime for Long-term Therapy...	NCT01398072	PHASE3	UNKNOWN	Chronic Obstructive Pulmonary Disease (COPD).

Recruitment Analysis

Overview of the current recruitment status across identified studies.

UNKNOWN: 7 studies
COMPLETED: 30 studies
ACTIVE_NOT_RECRUITING: 3 studies
WITHDRAWN: 4 studies
NOT_YET_RECRUITING: 3 studies
RECRUITING: 2 studies
TERMINATED: 1 studies

Intellectual Property & Legal

Core Patent Status

PRIMARY PATENT	EXPIRY
Pioneer/Historical Patent	1982

Patent Family Landscape

Claims Analysis

Composition of Matter: Determined by Expiry — *Broad synthesis claims*

Global Supply Chain

Registered Vendors (PubChem)

Publicly listed chemical vendors and suppliers:

VENDOR NAME	SOURCE TYPE	VERIFIED
Mylan	Major Global Supplier	**LLM Synthesized**
Pfizer	Major Global Supplier	**LLM Synthesized**
Sandoz	Major Global Supplier	**LLM Synthesized**
Teva Pharmaceutical Industries	Major Global Supplier	**LLM Synthesized**
Zydus Cadila	Major Global Supplier	**LLM Synthesized**

Strategic Conclusions

SWOT Analysis

Strengths

- Targeting matrix metallopeptidase 13
- Targeting matrix metallopeptidase 7

Weaknesses

- Requires competitive pricing analysis

Opportunities

- Repurposing potentials

Threats

- Regulatory changes

Final Recommendation

Given the current landscape, repurposing Doxycycline for new indications may be challenging due to its generic status and the resulting difficulty in differentiating it from existing products. However, the presence of two active clinical trials suggests that there may still be opportunities for innovation, particularly if new formulations or delivery methods can be developed that offer improved efficacy, safety, or convenience. To create a viable business strategy, it would be essential to focus on developing new IP, such as novel formulations or combination products, that can provide a competitive advantage in the market. Additionally, any repurposing efforts should be supported by robust clinical evidence and a thorough understanding of the target market and competition. While the existing publications and ongoing trials provide some evidence of Doxycycline's potential, it is crucial to approach any repurposing strategy with a critical and nuanced perspective, recognizing both the opportunities and challenges associated with this mature generic drug. Therefore, a potential strategic recommendation could be to explore the development of new formulations or combination products that leverage Doxycycline's existing efficacy and safety profile while creating new IP and competitive

Doxycycline's existing efficacy and safety profile while creating new IP and competitive advantages. This approach would require significant investment in research and development, as well as a thorough analysis of the market and competition, to determine the viability of such a strategy.

Approved By: _____

Chief Scientific Officer

Date: _____